

RIIC-ReID: Retrieval-Informed Illumination Correction for Cross-Light Re-Identification

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Abstract

Most illumination correction methods optimize images for human observers. They improve contrast, brightness, and visual naturalness in image space. Retrieval systems, however, are judged by a different observer: the downstream model. This creates a structural mismatch between human-perceptual enhancement and model-centric adaptation. We study this mismatch as a general problem in re-identification and ground it in wildlife re-identification, where cross-camera illumination shift is especially severe and identity often depends on local texture cues. We therefore formulate illumination handling not as standalone enhancement, but as constrained, retrieval-driven adaptation. Weak photometric priors define a feasible correction set, while retrieval geometry selects the preferred correction inside that set. We instantiate this view as RIIC-ReID, a retrieval-informed illumination correction framework that couples a compact illumination operator to the re-identification backbone through three principles: feasible correction, trust-controlled adaptation, and geometry-guided supervision. The framework keeps raw, safe base-corrected, and refined branches, and uses learnable trust gates to select among them per local region. On ATRW under a matched protocol, RIIC-ReID raises mAP from a baseline of 56.69 to 58.11 and remains stronger than an illumination-only front-end. Standardized fixed query-gallery experiments on GZGC Zebra and StripeSpotter show the same direction, while generic perceptual enhancers such as Zero-DCE++ and RetinexNet remain weaker or less consistent. These results support a simple conclusion: for retrieval, illumination correction should be optimized for the model rather than for human perception alone.

CCS Concepts

• **Computing methodologies** → **Image representations**; *Image processing*; • **Information systems** → **Retrieval models and ranking**.

Keywords

illumination correction, re-identification, retrieval, embedding geometry, task-aligned front-end

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1 Introduction

Most illumination correction methods are built for human observers. They improve contrast, brightness, and visual naturalness in image space [4, 13, 14, 21, 23, 24, 31, 36, 41, 45, 48, 50, 53, 57, 58, 61, 63]. That objective is sensible when the final consumer is a person, but in retrieval systems the final observer is not the human eye: it is the downstream model. This raises a more fundamental question

than low-light enhancement itself: *should illumination correction still be defined by human perception, or should it adapt to the feature space preferred by the model?*

This question is general across re-identification, but it becomes especially visible in wildlife settings, where capture conditions are less controlled and identity often depends on fine-grained local texture and species-specific pattern cues [5, 19, 42, 43]. In such cases, even strong modern encoders must absorb large camera- and illumination-induced appearance shifts [6, 9, 12, 16, 17, 34, 39, 67–69]. Two images of the same individual captured under different cameras or lighting conditions can differ more in brightness, local shadow, highlight pattern, and color cast than two images of different individuals captured under similar lighting. Once this happens, illumination handling is no longer a cosmetic preprocessing detail; it directly reshapes the feature geometry on which retrieval succeeds or fails. An image that looks more natural to a person need not be the image that best preserves identity-discriminative structure for a model (Figure 1).

This paper studies that mismatch explicitly. Our motivation is broad, but the empirical focus of this paper is cross-light re-identification. We do not propose a new physical theory of illumination; rather, we argue that illumination correction should be *observer-dependent*: human-centered enhancement and model-centered adaptation are related but different optimization problems. Retinex-style structure [23, 24] remains useful, but only as a weak admissibility prior [4, 48, 50, 53, 57, 63]. Once this distinction is made explicit, illumination handling should no longer be treated as an isolated enhancement block placed before recognition. It should instead be treated as a constrained and learnable front-end of the retrieval system itself [10, 15, 25, 30, 35, 49, 54]. In this view, photometric plausibility defines what corrections are admissible, and retrieval behavior determines which admissible correction is actually preferred. We validate this view primarily in wildlife re-identification, where illumination variation is a first-order nuisance and local texture makes correction errors especially costly; we use that setting as a stress-test domain rather than as a claim of universal retrieval coverage.

The challenge is that a purely task-driven corrector is ill-posed. This tension is already visible in ReID settings that either push illumination robustness into the descriptor [20, 26, 55, 56, 60] or introduce dedicated nighttime enhancement pipelines [32, 33]. If trained only through retrieval loss, the corrector may discover non-physical or dataset-specific transformations that reduce training error while harming generalization. If constrained too strongly by perceptual priors, it may never move samples toward the feature structure actually preferred by the encoder. The problem is therefore not simply to learn a stronger correction function, but to balance *task alignment*, *feasibility*, and *trust*. Viewed this way, the value of the method is not a new photometric law; it is an initial but concrete step toward model-adaptive illumination correction, in

which the front-end is trained for the downstream model rather than for human perception alone.

We address this problem with RIIC-ReID, a retrieval-informed illumination correction framework learned jointly with the downstream encoder. The framework is organized around three principles derived from the task formulation: *feasible correction*, which bounds the correction inside a safe photometric set; *trust-controlled adaptation*, which permits correction only where image-space and feature-space evidence justify it; and *geometry-guided supervision*, which lets retrieval structure rather than perceptual preference determine the final correction direction.

Contributions. (1) We introduce a general model-centric view of illumination correction for retrieval, where correction should follow the downstream model’s discriminative geometry and photometric priors act only as feasibility constraints. (2) We instantiate this view in RIIC-ReID, a constrained retrieval front-end that combines bounded inverse illumination, model-aware residual adaptation, identity protection, stripe-wise rollback, multi-branch fusion, and teacher-guided geometry. We study it primarily in wildlife re-identification, where illumination variation and texture-dependent identity cues are tightly coupled, making correction errors especially costly and the setting a natural stress test for feasibility and trust mechanisms. (3) We provide matched empirical evidence in that setting: on ATRW, naive illumination drops mAP from the baseline’s 56.69 to 55.62, while full RIIC-ReID reaches 58.11; the same ordering remains visible across backbones, standardized fixed query-gallery tests on GZGC Zebra and StripeSpotter, and perceptual baselines such as Zero-DCE++ and RetinexNet.

2 Related Work

2.1 Photometric and Retinex-Style Correction

Classical Retinex theory [23, 24] models an image as the product of reflectance and illumination, then recovers a reflectance-like representation by dividing out the estimated illumination. Deep Retinex methods, curve-estimation models, and recent unfolding, SNR-aware, structure-guided, transformer, diffusion, and implicit-representation variants learn this decomposition or related image-formation constraints end-to-end [4, 13, 14, 21, 31, 36, 41, 45, 48, 50, 53, 57, 58, 61, 63]. These methods can substantially improve visual quality, but their target remains photometric plausibility rather than retrieval utility.

2.2 Task-Aware Front-Ends for Recognition

Another line of work evaluates front-end processing through downstream utility rather than perceptual fidelity alone. In adverse-weather, day-night, and low-light recognition, task-conditioned front-ends adapt image processing to detection or tracking heads and improve downstream performance [10, 15, 25, 30, 35, 49, 54]. Related work also studies day-night transfer for retrieval itself [26]. Our work belongs to this task-aware family, but differs in making the mismatch between perceptual and retrieval optima explicit and using it to design both the front-end and its supervision.

2.3 Re-Identification Under Illumination Shift

Re-identification methods often absorb illumination variation indirectly through augmentation, stronger evaluation protocols, large-scale pre-training, and modern CNN/Transformer backbones [6, 9, 12, 16, 17, 34, 39, 40, 65, 67–69]. Several works address illumination more explicitly. Illumination-invariant ReID and Black ReID [20, 56] design representations intended to be more stable under difficult lighting, while later work studies domain generalization, relevance-aware experts, person-scale meta-generalization, exemplar-memory adaptation, clothing change, visible-infrared transfer, and explicit nighttime matching [3, 7, 22, 32, 33, 51, 55, 59, 60, 62, 64, 66]. Wildlife and animal re-identification has likewise expanded from tiger-specific pipelines to broader surveys, ecological analyses, toolkits, and multi-species resources [1, 2, 5, 11, 19, 27–29, 37, 38, 42, 43, 47]. Our view is complementary: instead of pursuing illumination robustness only in the descriptor or the dataset, we learn a bounded front-end whose correction is selected by downstream retrieval behavior. In this paper, we study that idea through a retrieval-centered formulation and use wildlife benchmarks as a stress-test domain where illumination errors are especially exposed.

3 Why Perceptual Illumination Correction Is Not Retrieval-Optimal

3.1 Task-Oriented Formulation

Let $x \in [0, 1]^{3 \times H \times W}$ denote an RGB image, y its identity label, g_ϕ a learnable illumination front-end, and f_θ the downstream retrieval encoder. We denote by

$$\tilde{x} = g_\phi(x), \quad z = f_\theta(\tilde{x}) \quad (1)$$

the corrected image and its embedding, respectively.

For retrieval-oriented illumination handling, we solve

$$\min_{\phi, \theta} \mathbb{E}_{(x, y)} [\mathcal{L}_{\text{task}}(f_\theta(g_\phi(x)), y)] \quad \text{s.t.} \quad g_\phi(x) \in \mathcal{F}(x), \quad (2)$$

where $\mathcal{F}(x)$ is a feasible set that enforces bounded gain, bounded chromatic drift, and identity-preserving deviations from the original image.

Equation (2) differs from conventional enhancement objectives [4, 14, 21, 50, 63], in that visual plausibility is enforced only through the feasible set $\mathcal{F}(x)$; the optimization objective itself is entirely retrieval-driven.

3.2 Why the Two Optima Generally Differ

To make the mismatch explicit, define the retrieval objective $L_x(\tilde{x}) = \mathcal{L}_{\text{task}}(f_\theta(\tilde{x}), y)$ and a photometric objective $P_x(\tilde{x})$ specified without reference to downstream retrieval geometry. Within the same feasible set $\mathcal{F}(x)$, consider three solutions:

$$\tilde{x}_P^* = \arg \min_{\tilde{x} \in \mathcal{F}(x)} P_x(\tilde{x}),$$

$$\tilde{x}_T^* = \arg \min_{\tilde{x} \in \mathcal{F}(x)} L_x(\tilde{x}),$$

$$\tilde{x}_\lambda^* = \arg \min_{\tilde{x} \in \mathcal{F}(x)} (L_x(\tilde{x}) + \lambda P_x(\tilde{x})).$$

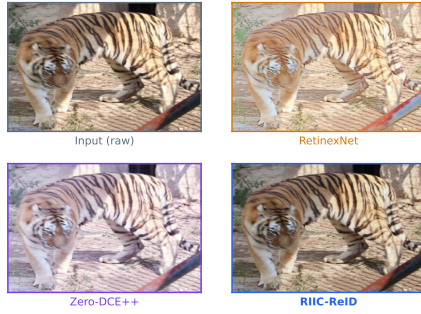
The first is the photometric optimum, the second the retrieval optimum, and the third the weak-prior solution used in our formulation.

The non-identifiability of Retinex-style decomposition already implies that some prior is unavoidable [4, 24, 50]: for any positive

(a) Retrieval results under different illumination corrections



(b) Illumination correction



(c) Embedding proximity: same ID vs. confuser

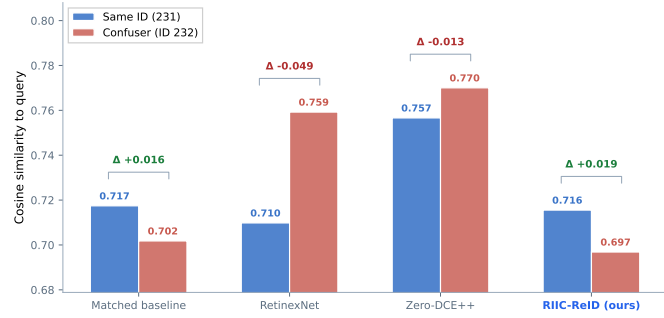


Figure 1: Retrieval comparison on a representative ATRW query (ID 231, luminance 0.57). (a) Top-5 gallery results under four illumination-handling strategies. Green borders denote correct matches; red borders denote wrong identities. RetinexNet and Zero-DCE++ visually improve the query but introduce confuser retrievals (ID 232), while RIIC-ReID returns all correct matches. (b) Correction comparison: the four query variants differ subtly in image space, yet produce markedly different retrieval outcomes. (c) Average cosine similarity between the query embedding and same-identity vs. confuser gallery embeddings. Perceptual methods invert the margin (query becomes closer to the confuser), whereas RIIC-ReID achieves the largest positive margin ($\Delta = +0.019$).

field c , one may write $x = (R \odot c) \odot (L/c)$. The relevant question is therefore not whether a prior exists, but whether it should dominate the objective or merely regularize the feasible set.

Prop. 1 (Local mismatch). Assume $\mathcal{F}(x)$ is an embedded C^1 feasible submanifold in a neighborhood of a photometric optimum $\tilde{x}_P^*$, and L_x, P_x are differentiable on that neighborhood. Let

$$\mathcal{T}_P = T_{\mathcal{F}(x)}(\tilde{x}_P^*), \quad g_P = \Pi_{\mathcal{T}_P} \nabla L_x(\tilde{x}_P^*).$$

If $g_P \neq 0$, then $\tilde{x}_P^*$ is not a local minimizer of L_x over $\mathcal{F}(x)$ and therefore cannot coincide with $\tilde{x}_T^*$.

Proof sketch (full proof in supplementary material). Since $g_P \in \mathcal{T}_P$, there exists a feasible C^1 curve $\gamma(t) \subset \mathcal{F}(x)$ with $\gamma(0) = \tilde{x}_P^*$ and $\gamma'(0) = -g_P$, so

$$\left. \frac{d}{dt} L_x(\gamma(t)) \right|_{t=0} = \langle \nabla L_x(\tilde{x}_P^*), -g_P \rangle = -\|g_P\|_2^2 < 0.$$

Hence L_x strictly decreases along a feasible direction through $\tilde{x}_P^*$, so the photometric optimum cannot be retrieval-optimal on $\mathcal{F}(x)$ unless the projected retrieval gradient vanishes.

Proposition 1 formalizes the key point of the paper. A perceptually reasonable correction is not retrieval-optimal unless the photometric optimum is also stationary for retrieval on the same feasible manifold. Such alignment is possible, but it is not guaranteed by Retinex-style admissibility alone. Conversely, the vanishing of the projected retrieval gradient is only a necessary local condition, not a sufficient one for the two optima to coincide.

3.3 A Local Justification for Weak Priors

We now justify why the photometric term should be weak. Introduce local coordinates δ on $\mathcal{F}(x)$ around a retrieval optimum $\tilde{x}_T^*$, so $\delta = 0$ corresponds to $\tilde{x}_T^*$. Assume $L_x(\delta)$ and $P_x(\delta)$ are twice continuously differentiable and that the restricted Hessian $H = \nabla^2 L_x(0)$

is positive definite on the feasible tangent space. Consider the weak-prior objective

$$J_\lambda(\delta) = L_x(\delta) + \lambda P_x(\delta), \quad \lambda \geq 0.$$

Prop. 2 (Weak-prior perturbation). Under the assumptions above, for sufficiently small λ there exists a local minimizer δ_λ^* of J_λ satisfying

$$\delta_\lambda^* = -\lambda H^{-1} \nabla P_x(0) + o(\lambda).$$

Consequently,

$$\|\delta_\lambda^*\|_2 = O(\lambda), \quad L_x(\delta_\lambda^*) - L_x(0) = O(\lambda^2).$$

Proof sketch (full proof in supplementary material). The stationarity condition is

$$\nabla L_x(\delta_\lambda^*) + \lambda \nabla P_x(\delta_\lambda^*) = 0.$$

Because $\delta = 0$ is a retrieval optimum, $\nabla L_x(0) = 0$. Since H is invertible, the implicit function theorem gives a C^1 local solution branch and therefore

$$\delta_\lambda^* = -\lambda H^{-1} \nabla P_x(0) + o(\lambda).$$

In particular, $\|\delta_\lambda^*\|_2 = O(\lambda)$. Substituting this into the second-order expansion of L_x around 0,

$$L_x(\delta) = L_x(0) + \frac{1}{2} \delta^\top H \delta + o(\|\delta\|_2^2),$$

yields the claimed $O(\lambda^2)$ excess retrieval risk.

Proposition 2 shows that a weak photometric prior perturbs the retrieval solution only to first order, while the induced retrieval risk increases only to second order. (The effective range of this regime depends on the condition number of H : an ill-conditioned Hessian amplifies $\|\delta_\lambda^*\|_2$ for a given λ , which motivates the conservative annealing schedule used in practice.) This is exactly the regime we want: photometric structure regularizes the local search space without dictating the correction target.

3.4 Design Implications

A final property completes the analysis: once model-aware adaptation moves beyond bounded photometric correction, the resulting change must remain trust-bounded.

Prop. 3 (Trust-bounded adaptation). Suppose the corrected image is produced by

$$\tilde{x} = \alpha \odot \tilde{x}_{\text{prot}} + (1 - \alpha) \odot x, \quad 0 \leq \alpha \leq 1,$$

and a stripe feature is fused by

$$F_{\text{fuse}}^{(s)} = \sum_{b=1}^B \alpha_b^{(s)} F_b^{(s)}, \quad \alpha_b^{(s)} \geq 0, \quad \sum_{b=1}^B \alpha_b^{(s)} = 1.$$

Then, for any monotone norm,

$$\|\tilde{x} - x\| \leq \|\tilde{x}_{\text{prot}} - x\|,$$

and, for each stripe s and any norm,

$$\|F_{\text{fuse}}^{(s)} - F_{\text{raw}}^{(s)}\| \leq \sum_{b \neq \text{raw}} \alpha_b^{(s)} \|F_b^{(s)} - F_{\text{raw}}^{(s)}\|.$$

Proof sketch (full proof in supplementary material). The image-space bound follows because $\tilde{x} - x = \alpha \odot (\tilde{x}_{\text{prot}} - x)$ with $0 \leq \alpha \leq 1$ elementwise. The feature-space bound follows because softmax fusion makes each fused stripe a convex combination of branch features, so

$$F_{\text{fuse}}^{(s)} - F_{\text{raw}}^{(s)} = \sum_{b \neq \text{raw}} \alpha_b^{(s)} (F_b^{(s)} - F_{\text{raw}}^{(s)}),$$

after which the triangle inequality gives the claim.

Proposition 3 explains why rollback and branch fusion are not heuristic extras: they implement explicit image-space and feature-space trust bounds.

The three propositions now yield a direct algorithmic reading. Proposition 1 says that photometric admissibility alone cannot determine the retrieval-preferred correction. Proposition 2 says that the photometric term should regularize the search only weakly. Proposition 3 says that the correction must remain trust-bounded in both image space and feature space. These observations lead to three requirements for a retrieval-oriented illumination front-end. First, correction must remain *feasible*: the model needs bounded gain, bounded chromatic drift, and weak physical regularity so the learned operator does not collapse into arbitrary image manipulation. Second, correction must be *trust-controlled*: the exact projection onto $\mathcal{F}(x)$ is intractable once semantic constraints are introduced, so the system needs learnable image-space and feature-space damping mechanisms. Third, correction must be *geometry-guided*: once the feasible set is defined, the preferred correction should be selected by retrieval geometry rather than by perceptual preference. RIIC-ReID is a direct implementation of these three requirements.

4 RIIC-ReID

4.1 Overview

Figure 2 gives an overview. The RIIC-ReID framework couples a compact illumination operator to the retrieval encoder without turning the system into a free-form enhancer. The illumination branch first constructs a feasible base correction, then applies model-aware adaptation and lightweight refinement under explicit trust control. The encoder simultaneously keeps a raw branch, so the final descriptor can be selected from raw, bounded base-corrected, and refined evidence rather than being forced to trust one corrected image. This organization mirrors the theory above and stays close to the broader idea of task-aware front-ends for recognition [15, 30, 54]: photometric priors delimit the admissible set, trust mechanisms keep adaptation conservative, and retrieval geometry determines which admissible correction is actually worth using.

4.2 Feasible Illumination Correction

We adopt the standard reflectance–illumination factorization as an approximate inductive bias [4, 23, 24, 50],

$$x \approx R \odot L, \quad (3)$$

where R is a reflectance-like term and L is an illumination field. The factorization is not treated as an exact physical claim; it simply provides structure for a learnable front-end.

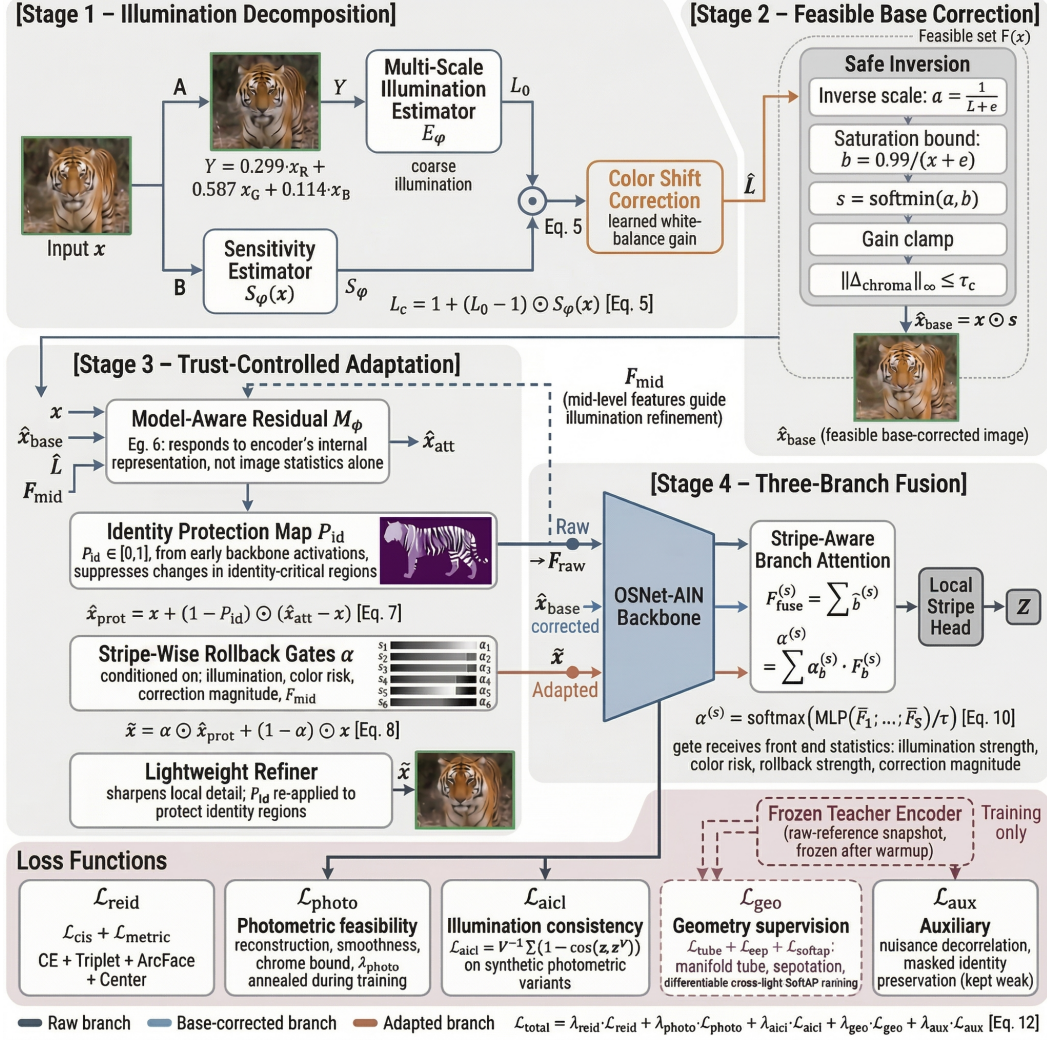


Figure 2: RIIC-ReID architecture. A bounded base corrector defines feasible illumination correction (§4.2). A model-aware residual branch adapts correction using mid-level encoder features, while identity protection and stripe-wise rollback suppress risky edits (§4.3). Stripe-aware competitive fusion selects among raw, base-corrected, and refined branches. Teacher-guided geometry losses supervise the corrected features to remain within same-identity manifold tubes (§4.4).

The framework first computes a luminance map

$$Y = 0.299x_R + 0.587x_G + 0.114x_B, \quad (4)$$

then predicts a coarse illumination field $L_0 = E_\phi(Y)$. To allow spatially varying task sensitivity, it also predicts a multiplicative sensitivity map $S_\phi(x)$ and forms

$$L_c = 1 + (L_0 - 1) \odot S_\phi(x). \quad (5)$$

This yields a stable low-frequency estimate without forcing the same correction strength everywhere.

To handle color shift, the front-end can further augment the scalar illumination with a learned white-balance gain or a learned multi-channel illumination field. We denote the resulting effective correction field by $\hat{L}$. Direct inverse illumination correction is unstable in dark or saturated regions, so the framework applies a bounded

scaling rule. We compute the desired inverse scale $a = 1/(\hat{L} + \epsilon)$ and the per-pixel saturation-safe upper bound $b = 0.99/(x + \epsilon)$. The actual scale is obtained through a differentiable soft minimum $s = \text{softmin}(a, b)$, then clamped to a bounded gain range together with a bound on chromatic drift $\|\Delta_{\text{chroma}}\|_\infty \leq \tau_c$. The resulting base-corrected image is $\hat{x}_{\text{base}} = x \odot s$. This stage implements the feasible-set part of the theory: it gives the model a learnable but explicitly bounded correction space. These constraints are not the final target; they only ensure that subsequent task-driven adaptation starts from a safe operating region.

4.3 Trust-Controlled Adaptation

Feasible correction alone is not enough, because even an admissible correction need not be trustworthy everywhere. The framework therefore introduces two layers of trust control.

First, a safe coarse guidance image is fed into the early layers of the backbone to extract a mid-level feature map F_{mid} . This feature map conditions a model-aware residual branch:

$$\tilde{x}_{\text{att}} = M_\phi(x, \tilde{x}_{\text{base}}, \hat{L}, F_{\text{mid}}). \quad (6)$$

The correction therefore responds to the encoder’s internal representation rather than to image statistics alone. This follows prior task-aware enhancement pipelines that use downstream semantics to guide low-level processing [15, 30, 54], but here the signal is explicitly retrieval-driven.

Second, the model restricts where and how strongly this residual may act. An identity protection map $P_{\text{id}} \in [0, 1]^{1 \times H \times W}$ is constructed from early backbone activations and used to suppress changes in identity-critical regions:

$$\tilde{x}_{\text{prot}} = x + (1 - P_{\text{id}}) \odot (\tilde{x}_{\text{att}} - x). \quad (7)$$

It then predicts a stripe-wise rollback gate α , conditioned on illumination, color risk, correction magnitude, and F_{mid} , so each horizontal body region can independently return toward the raw observation:

$$\tilde{x} = \alpha \odot \tilde{x}_{\text{prot}} + (1 - \alpha) \odot x. \quad (8)$$

This image-space rollback acts as a learned trust-region mechanism: risky regions receive smaller effective correction steps. A lightweight refiner then sharpens local detail around the rollback solution, and the identity protection mask is applied again so refinement cannot overwrite protected regions.

Trust control continues in feature space. The framework preserves a raw branch in the encoder and fuses raw, base-corrected, and refined branches using stripe-aware competitive attention. In the simplified two-branch form,

$$F_{\text{fuse}} = F_{\text{raw}} + w \odot (F_{\text{corr}} - F_{\text{raw}}). \quad (9)$$

The full framework generalizes this to three branches $b \in \{1, 2, 3\}$ (raw, base-corrected, refined), where per-stripe fusion weights are learned:

$$F_{\text{fuse}}^{(s)} = \sum_{b=1}^3 \alpha_b^{(s)} F_b^{(s)}, \quad \alpha^{(s)} = \text{softmax}(\text{MLP}([\bar{F}_1^{(s)}; \dots; \bar{F}_3^{(s)}]) / \tau), \quad (10)$$

where $\bar{F}_b^{(s)}$ is the average-pooled descriptor of branch b at stripe s . Equation (9) is the special case $B=2$ with $w=\alpha_{\text{corr}}^{(s)}$; in either form, the raw branch anchors the representation so that correction acts as a bounded residual. The gate also receives compact front-end statistics such as illumination strength, color risk, rollback strength, and correction magnitude, so trust in the corrected branch remains explicitly bounded by the state of the illumination module. This feature-space gate is the second trust mechanism: even if the image branch is corrected aggressively, the representation need not trust it unconditionally.

4.4 Geometry-Guided Training

The framework is trained with three groups of objectives. The first is the standard retrieval loss $\mathcal{L}_{\text{reid}} = \mathcal{L}_{\text{cls}} + \mathcal{L}_{\text{metric}}$, instantiated with cross-entropy, triplet [18], ArcFace [8], and center-loss terms [52]; related metric-learning formulations such as FaceNet and Circle Loss provide closely related geometric viewpoints [44, 46]. The second is a weak photometric feasibility term $\mathcal{L}_{\text{photo}}$, which keeps the correction inside the bounded set described above, together with an illumination-consistency term on synthetic photometric variants, $\mathcal{L}_{\text{iicl}} = V^{-1} \sum_v (1 - \cos(z, z^{(v)}))$.

The third group provides the key retrieval-oriented supervision beyond standard descriptor-level ReID objectives [8, 9, 18, 69]. The training code exposes warmup, optional illumination-only, and joint phases; the precise ATRW schedule is given in Section 5.1. After warmup, we freeze a snapshot of the raw-reference encoder as a teacher and use it to supervise the geometry of corrected features. A teacher-manifold tube loss constrains corrected features to remain inside a same-identity neighborhood:

$$\mathcal{L}_{\text{tube}} = [\|d^{\parallel}\| - r]_+^2 + \lambda_{\perp} \|d^{\perp}\|_2^2, \quad (11)$$

where $d^{\parallel}$ and $d^{\perp}$ are projections onto the teacher subspace and its complement. We further use a teacher-manifold separation term $\mathcal{L}_{\text{sep}}$ to maintain class boundaries and a differentiable cross-light SoftAP objective $\mathcal{L}_{\text{softap}}$ to preserve listwise retrieval ordering under illumination change. To reduce mini-batch bias, the teacher gallery can be augmented with a queue of frozen teacher descriptors from recent batches. In the final ATRW mainline, other active auxiliaries are kept weak and implementation-oriented: masked identity-image preservation and nuisance decorrelation. Together these terms specify not only whether correction helps, but which local retrieval geometry the corrected feature should preserve.

The full objective is

$$\mathcal{L}_{\text{total}} = \lambda_{\text{reid}} \mathcal{L}_{\text{reid}} + \lambda_{\text{photo}} \mathcal{L}_{\text{photo}} + \lambda_{\text{iicl}} \mathcal{L}_{\text{iicl}} + \lambda_{\text{geo}} \mathcal{L}_{\text{geo}} + \lambda_{\text{aux}} \mathcal{L}_{\text{aux}}, \quad (12)$$

where $\mathcal{L}_{\text{geo}} = \mathcal{L}_{\text{tube}} + \mathcal{L}_{\text{sep}} + \mathcal{L}_{\text{softap}}$ and $\mathcal{L}_{\text{aux}}$ bundles the weak nuisance-decorrelation and masked identity-image preservation terms. During joint training, λ_{photo} is annealed from a moderate initial value to a small residual. This schedule matches the theoretical view: photometric priors stabilize the early search, then progressively yield to retrieval geometry.

5 Experiments

5.1 Experimental Setup

We evaluate primarily on ATRW [27], an open-set wildlife re-identification benchmark, and report single-camera mAP, cross-camera mAP, and their mean (mmAP) as the main selection metric. We choose ATRW as the main testbed because it exposes the target problem most clearly: severe cross-camera illumination variation coupled with texture-dependent identity cues. The experiments test three points: whether generic photometric correction alone helps retrieval, whether trust-controlled adaptation recovers gains over that generic correction, and whether the same ordering remains visible across backbones, datasets, and human-oriented enhancement baselines.

Baseline. Unless otherwise stated, the main ablation uses OSNet-AIN [68]. We define a matched ReID baseline (hereafter: *baseline*) as the same backbone, stripe head, input resolution (256×384), PK sampler ($P=8$, $K=4$), ReID loss suite [8, 18, 34], and total training budget as the corresponding RIIC-ReID run, but without the illumination front-end or geometric supervision. For clarity, the matched main ablation contains four within-budget variants: *baseline*; *naive illumination*, which keeps only a safe correction branch; *illumination only*, which adds trust-controlled correction but no teacher geometry; and *full RIIC-ReID*, which enables all components. We additionally construct a generic illumination control that keeps a learned safe-correction branch but removes model-aware residual refinement, identity protection, stripe-wise rollback, raw/corrected branch fusion, and teacher-guided geometry. This isolates whether standalone illumination correction already helps retrieval, or only becomes useful after it is coupled to internal encoder evidence and retrieval geometry.

Training schedule. The code retains a three-phase training interface, but the final ATRW mainline collapses it into a practical two-stage run: a 15-epoch ReID warmup, a zero-epoch standalone illumination phase, and 66 joint-training epochs. Functionally, this is a warmup-plus-joint schedule with auxiliary weights ramped during the early joint stage; the teacher encoder for $\mathcal{L}_{geo}$ is frozen at the end of warmup.

Additional evidence. Beyond ATRW, we use standardized fixed query-gallery splits of GZGC Zebra and StripeSpotter as auxiliary cross-species robustness checks, in line with the broader move toward more explicit animal-ReID evaluation resources and benchmarking protocols [1, 5, 19]. These are not presented as official open-set benchmarks; final reported numbers come from fixed query/gallery splits, while checkpoint selection uses held-out train query/gallery splits. Their role is to test whether the same model-centric ordering survives beyond the main ATRW stress-test, not to claim a fully standardized universal suite for cross-species re-identification. We compare the same three variants as above: baseline, generic illumination, and full model. We also test whether human-oriented enhancement alone can explain the gain by pre-processing ATRW and GZGC Zebra with a Zero-DCE++ implementation and RetinexNet, then retraining the same OSNet-AIN baseline on the enhanced data.

5.2 Main Results

Table 1 presents the primary comparison. The pattern directly follows the logic of Propositions 1–3. The naive illumination control underperforms the baseline, with mmAP dropping from 56.69 to 55.62 and cross-camera mAP dropping from 40.87 to 39.60. This rules out the easy explanation that any learnable correction branch is sufficient: without trust control, correction can perturb identity-relevant structure on the hardest cross-camera metric.

Re-introducing task-aligned coupling reverses that behavior. The task-aligned illumination front-end raises mmAP to 57.86, which is +1.17 over the baseline and +2.24 over the naive control. This jump is the main evidence for trust-controlled adaptation: the problem is not illumination correction per se, but uncontrolled correction.

Table 1: ATRW main ablation under a matched protocol. Naive illumination removes model-centric trust and geometry coupling.

Variant	mAP single	mAP cross	mmAP	Rel. Baseline
Baseline	72.52	40.87	56.69	0.00
Naive illumination	71.65	39.60	55.62	-1.07
Illumination only	74.30	41.43	57.86	+1.17
Full RIIC-ReID	74.43	41.80	58.11	+1.42

Table 2: ATRW cross-backbone robustness under matched within-backbone protocols.

Backbone	Base mmAP	Full mmAP	Δ mmAP	Base Cross	Full Cross	Δ Cross
OSNet-AIN X1.0	56.69	58.11	+1.42	40.87	41.80	+0.93
OSNet X1.0	59.06	60.50	+1.44	43.35	44.75	+1.40
ResNet-50	56.61	57.62	+1.01	42.92	43.97	+1.05

Adding full teacher-guided geometry yields a smaller but still consistent further improvement to 58.11 mmAP, a +1.42 gain over the baseline and a +0.25 gain over the illumination-only front-end. The improvement is again concentrated on cross-camera mAP (41.80 for RIIC-ReID vs. 40.87 for the baseline and 39.60 for the naive control), which is the metric most exposed to illumination shift.

5.3 Cross-Backbone Robustness

The main table uses OSNet-AIN because it is the primary paper backbone, but our claim is not that RIIC-ReID is tied to one encoder family. To test this directly, we repeat the same baseline-versus-full-model comparison on OSNet-X1.0 and ResNet-50 while keeping the backbone-specific setup fixed within each row. Table 2 shows consistent positive gains: +1.42 mmAP on OSNet-AIN, +1.44 on OSNet-X1.0, and +1.01 on ResNet-50, with corresponding cross-camera mAP gains of +0.93, +1.40, and +1.05. The benefit therefore does not come from one backbone’s inductive bias alone.

5.4 Cross-Species Auxiliary Evidence

We next ask whether the same ordering survives beyond ATRW on fixed query-gallery evaluations with different species and capture conditions. Table 3 reports Rank-1 and mAP on GZGC Zebra and StripeSpotter. On GZGC Zebra, generic illumination already gives a moderate gain (72.71 vs. 70.32), but the full model improves further to 74.22. On StripeSpotter, generic illumination is slightly worse than the baseline (91.29 vs. 91.55), while the full model still reaches 93.89. This asymmetry is informative: generic correction is not uniformly harmful, but its effect is unstable, whereas the full task-aligned model remains best on both datasets.

5.5 Generic Perceptual Enhancement Baselines

We finally test whether the gain can be explained simply by better-looking inputs produced outside the retrieval model. Table 4 shows that on ATRW, Zero-DCE++ reaches 57.75 mmAP while RetinexNet

Table 3: Cross-species fixed query-gallery evidence on standardized splits. Model selection uses held-out train splits.

Dataset	Variant	Rank-1	mAP	Rel. Baseline
GZGC Zebra	Baseline	71.70	70.32	0.00
GZGC Zebra	Generic illumination	74.48	72.71	+2.39
GZGC Zebra	Full model	75.92	74.22	+3.90
StripeSpotter	Baseline	93.00	91.55	0.00
StripeSpotter	Generic illumination	92.00	91.29	-0.26
StripeSpotter	Full model	94.00	93.89	+2.34

Table 4: Generic human-perceptual enhancement baselines with a task-aligned full-model reference. Dashes denote metrics undefined under the respective protocol: ATRW’s open-set evaluation does not define Rank-1 or overall mAP; GZGC Zebra uses a single-partition gallery without the single-camera/cross-camera distinction.

Dataset	Method	Rank-1	mAP	mAP single	mAP cross	mmAP
ATRW	Zero-DCE++	–	–	73.14	42.36	57.75
ATRW	RetinexNet	–	–	71.70	41.64	56.67
ATRW	Full RIIC-ReID	–	–	74.43	41.80	58.11
GZGC Zebra	Zero-DCE++	68.01	67.31	–	–	–
GZGC Zebra	RetinexNet	68.64	68.51	–	–	–
GZGC Zebra	Full RIIC-ReID	75.92	74.22	–	–	–

stays at 56.67; both remain below the full model at 58.11. On GZGC Zebra, Zero-DCE++ and RetinexNet reach 67.31 and 68.51 mAP, again well below the task-aligned full model at 74.22. Generic enhancement is therefore not useless, but it is weaker and less reliable than retrieval-aligned correction.

5.6 Interpretation

Taken together, the results support a reliability claim rather than a universal dominance claim. Under a wildlife-focused evaluation suite, the generic-control row shows that learnable photometric correction is not automatically retrieval-beneficial; on ATRW it hurts the hardest metric. Once trust control and geometry are restored, bounded correction becomes useful and remains the most stable variant across backbones, cross-species splits, and perceptual baselines.

This is consistent with the logic of Section 3: the generic-control failure is the empirical counterpart of Proposition 1, the weaker, less stable behavior of generic enhancement is consistent with Proposition 2, and the gain from rollback and branch fusion matches

Proposition 3. The stronger claim is therefore not that every non-task-aware corrector fails, but that generic correction is unstable while retrieval-aligned correction is more reliable.

6 Limitations

Our formulation is intentionally conservative. Section 3 gives a local feasible-manifold analysis under smoothness and local curvature assumptions, not a proof of global optimality for deep retrieval training. Although the formulation is general, our strongest empirical validation is currently concentrated on wildlife benchmarks, where illumination shift is a salient nuisance; broader human-centered validation remains future work. RIIC-ReID also remains most appropriate when illumination is a major nuisance and identity depends on local texture structure; if lighting is not the bottleneck, or if a much larger backbone can already absorb it, the utility of a structured lighting with minimal illumination variation—the rollback gates and trust mechanisms are designed to converge toward the raw branch, but we have not empirically verified that the front-end introduces zero residual perturbation in such benign settings. Finally, RIIC-ReID is architecturally heavier than the baseline, and we do not yet provide a dedicated runtime, parameter, or memory breakdown, so its benefit must be judged against deployment cost.

7 Conclusion

We presented RIIC-ReID, a task-aligned illumination correction framework for cross-light re-identification. The central claim is motivated by retrieval more broadly: the front-end should be optimized for retrieval geometry rather than perceptual preference alone. We instantiate and primarily validate that view in wildlife re-identification, where cross-camera illumination shift is especially severe. Weak photometric priors define a feasible set, trust mechanisms regulate how far correction is allowed to move, and teacher-guided geometry determines which direction is worthwhile inside that set. Proposition 1 motivates task alignment over pure photometric optimization, Proposition 2 motivates weak photometric regularization, and Proposition 3 motivates rollback and trust-bounded fusion.

Empirically, the contrast between RIIC-ReID and the generic illumination control shows that illumination correction is not inherently beneficial for retrieval, but becomes useful and reliable once it is constrained by feasibility, moderated by trust, and selected by downstream retrieval structure. Across ATRW, cross-backbone tests, standardized fixed query-gallery evaluations on GZGC Zebra and StripeSpotter, and perceptual baselines such as Zero-DCE++ and RetinexNet, the ordering is consistent: the most reliable gains come from learning illumination handling with the retrieval model rather than designing it for human perception alone.

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